Beautified Is Not Forged: Measuring and Separating Beautification from Face Forgery

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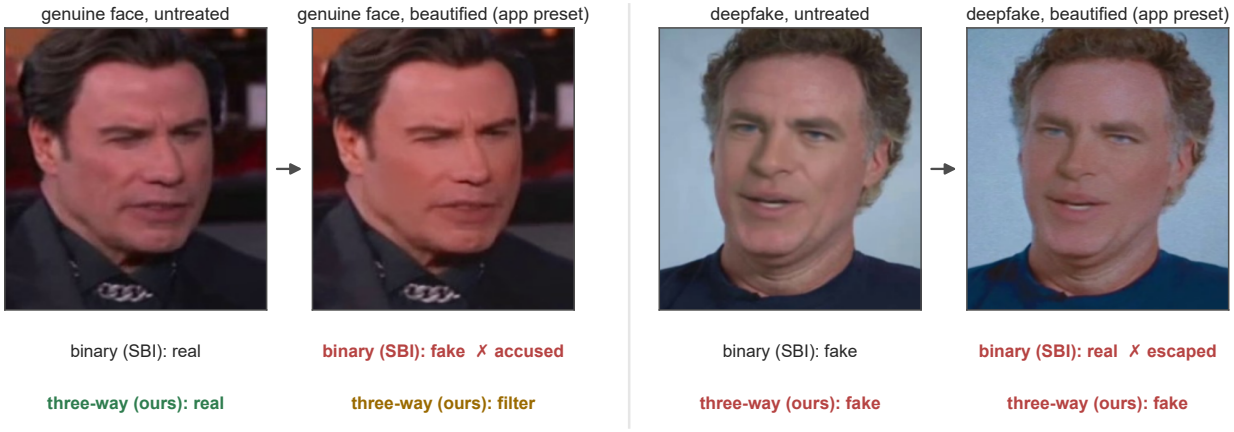


Fig. 1. Celeb-DF-B frames of a genuine video and of a deepfake, before and after the same app beautification. The official SBI checkpoint, at the threshold that accuses 5% of untreated genuine frames, accuses the beautified genuine face and lets the beautified deepfake pass. Ours calls the first *filter* and the second *fake*. Frames were chosen where SBI errs after beautification; rates over the benchmark are in Table II.

***Abstract*—Many of the faces a forgery detector receives have been beautified by a camera app or a retouching service. We measure what this does to fifteen published detectors with one rule for all of them: the score is the probability of forgery, the threshold is set on untreated genuine faces, and both errors are counted, genuine faces accused and forgeries let through. On three sources, app presets on Celeb-DF-B, eleven colour filters on Celeb-DF-v2 and sixteen further filters on DFD, every detector that detects the untreated forgeries loses AUC under app presets; the four CLIP-based detectors of 2025–26 lose 0.08 to 0.12, and a retouching service makes eleven detectors accuse 36–63% of its most strongly smoothed genuine faces. A probe of the feature space shows where the damage comes from: a radial luminance overlay moves a genuine face along the forgery direction three to ten times further than a pure colour look, and the direction cannot be projected out without losing the forgery evidence itself. A controlled ablation over three seeds shows that the sign of the error is set in training: a detector shown beautified genuine faces only lets 21 points more beautified deepfakes through, more than one shown no beautification at all. With beautification on both classes, commercial retouching pairs split into single operations, and part-level forgeries with exact footprints, a CLIP ViT-L/14 model with LoRA keeps an AUC of 0.87–0.88 under app presets against 0.81 and 0.83 for Effort and Forensics Adapter, is on par with both on untreated Celeb-DF-v2 and DFD under each detector's own preprocessing, calls 0.6% of an unseen service's retouched faces fake, and names regions whose restoration removes 94% of its detections on held-out part-level forgeries, within a point of the true footprint. Seven variants of adversarial training over a measured beautification operator, a learned inverse, and a decision read from the evidence map did not improve on this; we report them so that others need not repeat them.**

I. Introduction

Face-forgery detectors are used for content moderation, remote identity checks and fact-checking. In these settings a genuine face called fake accuses a real person, and a forgery called real is let through. Many of the genuine faces such a detector receives have been beautified. Retouching is common on social media and photo-sharing sites and appears even on identification documents [1]; skin smoothing, whitening, eye enlargement and face slimming are default options in camera apps and are sold by retouching APIs, and colour presets ship with every photo-sharing app. Beautification edits the pixels of the face, which is what a forgery detector inspects, but changes neither who is shown nor what the person does.

We measure both errors on fifteen published detectors, from Xception [2] to the CLIP-based Effort [3], Forensics Adapter [4], GenD [5] and DFD-HR [6], with one rule for every detector: the score is its probability of forgery, the threshold is the one that accuses 5% of untreated genuine frames, and a forgery not called fake is a miss whatever else it was called. Three test sources carry beautification that none of the detectors was trained on. On Celeb-DF-B [7], whose app presets were applied by its authors, re-encoding alone explains part of the change, so we subtract it; the miss rate still rises by 11.0, 15.2 and 21.8 points for Effort, Forensics Adapter and SBI [8]. On Celeb-DF-v2 frames under eleven colour filters

never used in our training, the AUC of the two CLIP detectors falls from 0.88–0.89 to 0.81. On DFD frames under sixteen further filters, built after every method decision of this paper had been taken, it falls from 0.94 to 0.90–0.92 and Effort's miss rate rises by 15 points. The other direction is as real: at a threshold that accuses 5% of a retouching service's untouched originals, the eleven FF++-trained detectors accuse 36–63% of its most strongly smoothed genuine faces.

Where does the damage come from? We look at the pooled feature of a CLIP detector trained on FF++ only. Of eleven held-out filters, the two that add a radial luminance overlay move a genuine face's forgery logit by $+5.6$ and $+7.8$ on average, the nine pure colour looks by $+0.1$ to $+2.4$, and the overlay shifts have a cosine of 0.20–0.33 with the forgery weight vector against 0.00–0.09 for the colour looks. A luminance ramp around the face reads as a blending boundary, which is the published mechanism of frame-based detectors [9]. Most of the shift energy of every held-out filter lies in the subspace spanned by the shifts of beautifications seen in training, so the directions were seen and the combinations were not. Projecting that subspace out and refitting the head does not help: the clean AUC drops at once and the false accusations under the filters do not fall, because the same few directions carry the forgery evidence.

What a detector learns about beautification is therefore set in training, and we test how. With the architecture, data and recipe held fixed, we vary only which class receives beautified training views. Beautifying genuine faces only, the recipe a "filter-robust real class" would suggest and the direction of fine-tuning on enhanced samples [10], reliably makes beautified deepfakes escape more than showing no beautification at all (21 against 11–17 points of miss rise, three seeds, two tuning schemes); beautifying both classes is on average the best setting. This contradicts the rule of augmenting only the forgery side [11] and explains the sign of the errors we measure on published detectors.

Our contributions are:

- **A two-sided measurement with one rule.** Fifteen detectors on three beautification sources and a commercial retouching service, scored by the same rule, with video-level confidence intervals, re-encoding separated from beautification, and each baseline also run under its own preprocessing.
- **The mechanism, measured.** A feature-space probe that quantifies which edits move a CLIP detector toward "forged", by how much, and why a linear fix cannot work.
- **Exposure decides the sign of the error.** A three-arm ablation over three seeds and two tuning schemes showing that beautification shown to one class is learnt as a label, and that exposure moves the errors between misses and false alarms without changing separability.
- **A detector and an explanation that hold up.** With beautification on both classes, commercial retouching pairs split into single operations and part-level forgeries with exact footprints, a CLIP model with LoRA is on par with Effort and Forensics Adapter on untreated data, keeps 0.04–0.07 more AUC under app presets, and names regions whose restoration removes its detections.

We also report what did not work, because it bounds the claims: seven variants of adversarial training over a measured beautification operator, a learned inverse of the edit, and a decision read from the evidence map. A third output class for beautified faces, which we proposed in an earlier version of this work, does not improve detection either; a detector with the same data and a single forged output is better by about 0.02 AUC, and the class is kept as a user-facing output.

II. Related Work

Beautification and forgery detection. Libourel et al. [7] released Celeb-DF-B and measured AUC drops for three detectors under app presets; Saeed et al. [10] show that face enhancement lets deepfakes evade six detectors and fine-tune on fixed enhanced samples; Concas et al. [12] report both errors for two CNNs under a smoothing filter. None trains against the edit, and none scores detectors by a common rule. A NeurIPS 2025 benchmark of screen-recapture distortions [13] names beautification filters as future work. The retouching literature labels commercial operations [14] or restores the untouched face [15]–[18] without a forgery class, and Liu et al. [19] attribute both forgery and beautification tools in a closed set.

Attacks on forgery detectors through natural appearance. StatAttack [20] optimises exposure, blur and noise, and AdvShadow [21] a shadow overlay, to make forgeries pass; both are attacks in one direction. Fatima et al. [22] train on the worst of K random counter-forensic transforms. Adversarial training over semantic perturbations exists for image classification [23]–[26]; we tried its forgery-detection form with a measured beautification operator and report that it does not beat random exposure (Sec. V-D).

Generalisable forgery detection. Face X-ray [27] and SBI [8] synthesise blending artefacts from real images, UCF [28] separates common from method-specific forgery features, LSDA [11] augments the forgery side of the latent space, and Effort [3] and Forensics Adapter [4] adapt CLIP ViT-L/14 with few trainable parameters. DeepfakeBench [29] provides common checkpoints and the preprocessing these detectors were trained with. The alpha-blending hypothesis [9] holds that frame-based detectors localise compositing artefacts; our measurement that a luminance ramp around a genuine face raises its forgery score is its benign-edit counterpart.

Explanations for detectors. GAIN [30] trains attention so that erasing the attended region removes the evidence, amortised maskers predict saliency in one pass [31], [32], and DOLOS [33] compares methods for localising inpainted regions. We supervise the map with exact footprints and test it by restoring pixels.

III. Problem and Evaluation Rule

A face image is *forged* if it shows a person or an action that was not captured: an identity swap, a reenactment, a synthesised face, or a part replaced by another person's or by generated content. It is *beautified* if its appearance was

TABLE I
CASES NEAR THE CLASS BOUNDARY AND WHETHER THEY OCCUR IN OUR DATA.

Case	Forged?	In our data
Camera processing, JPEG, video coding	no	all sets
Make-up worn at capture	no	incidental
Beautification by an app or a service, neural or not	no (beautified)	scripted, 3 services, app presets
Reenactment by another person's performance	yes	Face2Face, NeuralTextures
A part from another face, or inpainted	yes	part-level forgeries
Face swap	yes	FF++, Celeb-DF, DFD
Beautified forgery	yes (and beautified)	FF++, Celeb-DF-B
Expression editing; synthesised face	yes	not tested

Algorithm 1 Single-operation components of a commercial render

Require: original O, render R, eye mask E, σ, noise floors τ_k

$f \leftarrow \mathrm{DIS}\big(\mathrm{LCN}(O), \mathrm{LCN}(R)\big)$
$P \leftarrow R \circ (\mathrm{id} - f^{-1}) - O$
$C_{\text{eye}} \leftarrow \mathcal{W}(O; E \odot f)$; $C_{\text{contour}} \leftarrow \mathcal{W}(O; (1-E) \odot f)$
$C_{\text{tone}} \leftarrow O + G_\sigma * P$; $C_{\text{texture}} \leftarrow O + P - G_\sigma * P$
for $k \in \{\text{eye}, \text{contour}, \text{tone}, \text{texture}\}$ **do**
  $q_k \leftarrow \text{Measure}_k(C_k, O)$; **if** $q_k \geq \tau_k$ **keep** $(C_k,\ e_k,\ M(C_k))$
end for

edited after capture while identity, expression, pose and speech stay those of the capture: colour and tone, skin texture, or a smooth reshaping of the person's own features. The two attributes are independent; a beautified forgery is forged. The tool does not decide: a retouching service that uses a neural network beautifies, and a reenactment that keeps the identity forges. Table I lists cases near the boundary. A detector has to return a forgery score that does not move when a face is beautified; the beautified attribute may be reported separately for the user.

One rule for every detector. Each detector is scored by its probability of forgery. On every source we set the threshold at a false-alarm rate of 5% on the untreated genuine frames, keep it fixed, and report under beautification the false-alarm rate FA (genuine faces called fake) and the miss rate MISS (forgeries not called fake), with the AUC of untreated and of beautified frames. A forgery that a three-way model sends to its beautified class is a miss; an earlier version of this work counted only forgeries called *real*, which hid 19.5% of the beautified Celeb-DF-B deepfakes that our model sent to the third class, and that rule is withdrawn. Confidence intervals are 95% percentile intervals over 1,000 (Celeb-DF-B) or 500 bootstrap resamples of videos; differences between detectors use identical resamples.

Sources. Celeb-DF-B [7]: 231 Celeb-DF-v2 videos, each as recorded, re-encoded, and beautified by one of four app presets, for genuine and synthesised faces (32 frames per video). The rise in MISS attributed to beautification is the rise under the preset minus the rise under re-encoding alone. Celeb-DF-v2 filters: the 516 test videos of Celeb-DF-v2 [34], 32 frames each, every frame also saved under one of eleven colour filters from an open filter library [35] that no model of ours trained on, assigned per video and applied to the full frame before the face crop. DFD filters: the 332 DFD videos [36], 32 frames each, under one of sixteen filters from a second library, grayscale and sepia, none used in training or in the diagnosis that motivated any design choice; the set was built after the method was fixed. Commercial retouching: 2,210 FFHQ originals and their renders by a service never seen in training (Sec. V-E). Untreated cross-dataset detection: Celeb-DF-v2 and DFD, each baseline also scored on the crops of its own training pipeline (dlib, five-point alignment, scale 1.3, 256 px [29]).

IV. METHOD

A. Supervision from paired edits

Every edited image in our training pool has a pixel-aligned original x_0, so the footprint of the edit is known exactly, $M(x) = \text{dilate}_\rho(\mathbf{1}[\|x - x_0\|_\infty > \tau])$ with $\tau = 6$ on 8-bit intensities and ρ the feather width.

FF++ triplets. FF++ c23 [37] under the official 720/140/140 video split. For each genuine training frame x_0 a step draws a genuine item, a beautified item (one of four landmark and skin operations at a random strength, one of 14 colour filters [35], or a random colour grading) and a forged item (a self-blend of x_0 [8], an observed forgery from the four FF++ methods, or a part-level forgery), and the forged item is beautified on top with probability 0.5 in the same way. The same random blur or downscale is applied to all three items, so that sharpness does not predict the label. The FF++-only models of Sec. V-C use nothing else.

Commercial renders. 7,831 FFHQ photographs [38] and their renders by two retouching services from RetouchingFFHQ [14], Tencent and Megvii, which apply eye enlargement, face lifting, whitening and smoothing together. Each render R of an original O is split into single-operation components in O's geometry: dense optical flow [39] on locally normalised luminance gives the warp f, the photometric residual is $P = R \circ (\mathrm{id} - f^{-1}) - O$, and the components are the eye and contour warps of O by the masked flow and the low- and high-frequency parts of P added to O (Alg. 1); applying the four in sequence reproduces R at 108 dB PSNR. Components whose measured edit falls under the noise floor are dropped, leaving 50,112 of 73,952. The photometric residual of a render is also added to a self-blend of the same original, so that a measured commercial edit appears on forged as well as on genuine faces. The third service, Alibaba, is used only for testing.

Part-level forgeries. 61,510 training forgeries on FF++ frames transplant the eyes, nose or mouth of another identity or regenerate the part with Stable Diffusion inpainting; pixels outside the feathered mask are copied from x_0, so the footprint

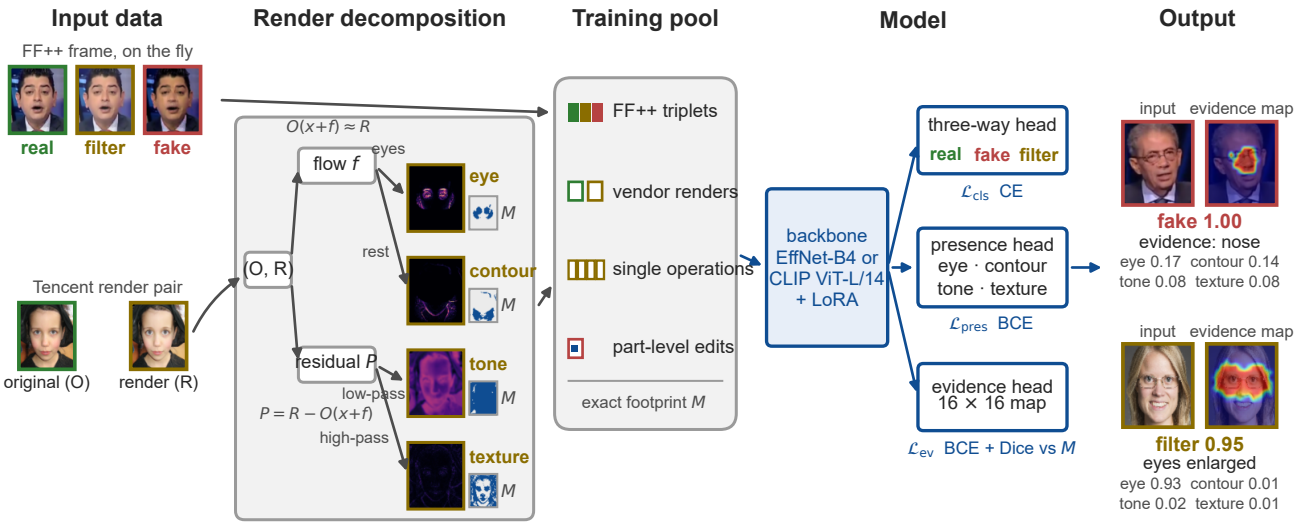


Fig. 2. Overview. Each genuine FF++ photograph yields a real / filter / fake triplet on the fly, with beautification drawn for the genuine and for the forged item; a beautified forgery stays *fake*. Each commercial render R of an original O is split by optical flow f and photometric residual P into single-operation components with exact footprints M (Alg. 1). One backbone feeds the forged and beautified logits, an operation-presence head and an evidence head trained against M. Right: outputs on a held-out part-level forgery and on a render from the unseen service.

is exact. The held-out test set uses test-split frames and donors from test videos only.

B. Model

The detector is a frozen CLIP ViT-L/14 [40] with rank-16 LoRA [41] on every attention projection (3.2 M trained parameters of 306 M). Two linear heads on the pooled feature give the forged logit z_f and the beautified logit z_b; the three-way output is the product of the two Bernoullis, $P(\text{real}) = \bar{\sigma}_f \bar{\sigma}_b$, $P(\text{fake}) = \sigma_f$, $P(\text{filter}) = \bar{\sigma}_f \sigma_b$, so the forgery score is $\sigma(z_f)$ whatever the beautified logit says. An operation-presence head and a 1×1 convolution on the spatial tokens (the evidence map $\hat{m}$, 16×16) are trained against the operation labels and the footprints M:

$$\mathcal{L} = \text{BCE}(z_f, y_f) + \text{BCE}(z_b, y_b) + \lambda_s \, \mathbf{1}[y_b] \, \text{BCE}(\hat{s}, s) + \lambda_m \, \mathbf{1}[M \text{ exact}] \left(\text{BCE}(\hat{m}, M) + \text{Dice}(\hat{m}, M)\right), \quad (1)$$

with $\lambda_s = \lambda_m = 1$. The classifier does not read $\hat{m}$, so whether the map explains the decision has to be tested (Sec. V-G). Training: 224 px crops, AdamW at 10^{-4} with warm-up and cosine decay, 8 epochs, a batch of 16 genuine bases with their forged and beautified items. The checkpoint is chosen on validation data only (FF++ validation macro-F1 for FF++-only models; for full-data models its mean with balanced accuracy on held-out originals and renders of the training services). No test source is used for any choice. A control with the same data and only the forged head ("binary") is trained the same way with validation AUC as the selection rule.

V. Results

A. Beautification hurts every detector, in both directions

Table II scores the fifteen detectors and ours by the rule of Sec. III. Every published detector loses AUC under beautification on every source, and the loss is largest for the detectors that detect best. On Celeb-DF-B the AUC of SBI falls from 0.875 to 0.752, of Effort from 0.909 to 0.807 and of Forensics Adapter from 0.909 to 0.828, and after re-encoding is subtracted the presets alone raise their miss rates by 21.8, 11.0 and 15.2 points; for the thirteen detectors that detect Celeb-DF-B deepfakes above chance the rise is 3.4 to 21.8 points, every interval above zero. GenD and DFD-HR, the two most recent, fall from 0.924 to 0.807 and from 0.938 to 0.838. The same two CLIP detectors fall from 0.88–0.89 to 0.81 under the Celeb-DF-v2 filters, where their miss rates rise by 23.3 and 11.9 points, and from 0.94 to 0.90–0.92 under the DFD filters, where Effort's miss rate rises by 15.1 points [11.1, 18.6]. The false-alarm side is small for these two on colour filters and large for UnivFD (+11.6 on DFD) and NPR, whose scores collapse under any filter. On commercial retouching the false alarms dominate: at a threshold that accuses 5% of the untouched originals, the eleven FF++-trained detectors accuse 36–63% of the most strongly smoothed genuine faces (Sec. V-E).

Two details matter for reading the table. Where the filter is applied matters: with the filter on the face crop rather than on the full frame, a vignette is centred on the face and Forensics Adapter loses 0.017 more AUC on DFD; we report the full-frame version and keep the crop-level one in the supplement. And a fixed threshold sees what AUC cannot: our full-data three-way model keeps its AUC under the DFD filters (0.942

TABLE II

BEAUTIFICATION HURTS EVERY DETECTOR IN BOTH DIRECTIONS. ONE RULE FOR ALL: SCORE $P(\text{fake})$, THRESHOLD AT 5% FPR ON UNTREATED GENUINE FRAMES. CELEB-DF-B: $\Delta\text{MISS}_{\text{beauty}}$ IS THE RISE IN MISSES UNDER THE APP PRESET MINUS THE RISE UNDER RE-ENCODING ALONE, FA_b THE FALSE-ALARM RATE ON BEAUTIFIED GENUINE FRAMES. CELEB-DF-V2 AND DFD: ELEVEN AND SIXTEEN COLOUR FILTERS NEVER USED IN TRAINING, APPLIED TO THE FULL FRAME; ΔMISS AND ΔFA IN PERCENTAGE POINTS WITH 95% VIDEO-CLUSTER INTERVALS. OURS ROWS: FF++-ONLY MODELS AND THE FULL-DATA MODELS OF SEC. V-C–V-F.

	Celeb-DF-B (app presets)				Celeb-DF-v2 + 11 held-out filters				DFD + 16 fresh filters			
Detector	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	FA_b	AUC	AUC_f	ΔMISS	ΔFA	AUC	AUC_f	ΔMISS	ΔFA
Xception	0.811	0.737	+10.9 [7.4, 14.1]	5.7	0.718	0.693	+1.1 [-1.0, 3.1]	+1.5 [-0.2, 3.2]	0.854	0.834	+2.0 [-0.6, 5.4]	+2.0 [0.2, 4.3]
EfficientNet-B4	0.811	0.762	+3.4 [0.9, 5.9]	7.8	0.750	0.725	+2.5 [0.2, 4.8]	+0.8 [-0.3, 2.6]	0.810	0.781	+6.3 [3.6, 9.2]	-0.5 [-1.9, 1.7]
F3Net	0.798	0.731	+10.9 [7.4, 14.8]	5.4	0.708	0.688	-0.6 [-2.7, 1.6]	+1.3 [-0.8, 3.8]	0.781	0.774	-8.5 [-11.5, -5.5]	+3.6 [2.1, 5.7]
SPSL	0.781	0.727	+12.4 [9.3, 15.5]	3.4	0.703	0.683	+3.1 [1.5, 4.9]	-0.2 [-1.5, 1.4]	0.829	0.837	+4.7 [3.2, 6.7]	-1.8 [-2.4, -0.9]
UCF	0.841	0.777	+11.3 [7.6, 15.4]	5.7	0.744	0.697	+3.1 [1.0, 5.5]	+0.9 [-0.6, 2.3]	0.822	0.804	+5.0 [1.9, 8.0]	+0.3 [-1.6, 2.5]
RECCE	0.785	0.723	+13.8 [10.7, 17.1]	4.1	0.688	0.661	+0.1 [-1.9, 2.0]	+0.9 [-0.8, 2.4]	0.833	0.808	+7.3 [4.6, 10.7]	-0.2 [-1.7, 1.3]
CORE	0.821	0.752	+13.5 [10.2, 16.6]	4.0	0.715	0.693	-0.0 [-1.5, 1.9]	+0.4 [-0.8, 2.4]	0.827	0.821	-5.4 [-8.8, -2.9]	+1.6 [-0.3, 3.7]
SRM	0.789	0.733	+15.3 [12.1, 18.7]	2.8	0.717	0.712	-0.1 [-1.4, 1.6]	+0.7 [-0.5, 2.1]	0.830	0.822	+5.8 [2.4, 9.7]	+0.2 [-1.2, 1.9]
SBI [8]	0.875	0.752	+21.8 [17.3, 26.1]	1.8	0.817	0.776	+8.2 [5.5, 10.4]	-0.6 [-1.5, 0.5]	0.829	0.810	+4.5 [2.7, 6.3]	+0.4 [-0.7, 1.4]
UnivFD	0.653	0.624	-7.2 [-10.2, -4.9]	7.6	0.662	0.622	-7.2 [-10.6, -4.2]	+7.8 [4.1, 11.7]	0.840	0.801	-10.2 [-13.5, -6.8]	+11.6 [8.0, 15.7]
NPR	0.503	0.498	-0.7 [-2.2, 0.4]	3.4	0.522	0.497	-38.0 [-43.2, -32.2]	+36.0 [28.7, 43.6]	0.473	0.516	-13.7 [-20.2, -8.4]	+9.4 [2.4, 15.3]
Effort [3]	0.909	0.807	+11.0 [7.7, 14.5]	5.4	0.881	0.813	+23.3 [19.9, 26.3]	-2.8 [-3.7, -1.7]	0.939	0.905	+15.1 [11.1, 18.6]	-2.3 [-3.1, -1.4]
Forensics Adapter [4]	0.909	0.828	+15.2 [11.9, 18.6]	7.9	0.887	0.814	+11.9 [8.6, 14.7]	+2.5 [0.3, 5.7]	0.940	0.924	+8.0 [5.6, 11.2]	-0.6 [-1.6, 0.4]
GenD [5]	0.924	0.807	+12.0 [8.2, 15.6]	9.9	0.891	0.856	-10.1 [-13.1, -7.1]	+9.9 [7.4, 13.1]	0.934	0.925	+3.1 [0.7, 5.8]	+1.2 [0.1, 2.2]
DFD-HR [6]	0.938	0.838	+3.6 [0.6, 6.8]	14.6	0.879	0.824	+8.0 [5.3, 10.5]	+1.6 [0.4, 3.8]	0.943	0.928	+9.1 [6.8, 11.8]	-2.0 [-2.9, -1.2]
OURS, FF++ only	0.938	0.872	+7.5 [4.8, 10.5]	7.8	0.898	0.851	-3.5 [-6.4, -0.9]	+7.2 [4.6, 10.6]	0.943	0.932	-0.3 [-2.6, 2.6]	+3.4 [2.1, 4.9]
OURS (full data, three-way)	0.938	0.866	+6.7 [4.0, 9.5]	9.5	0.891	0.873	+7.3 [4.5, 9.8]	-0.4 [-1.5, 0.8]	0.942	0.931	+8.9 [6.0, 12.2]	-1.0 [-2.4, 0.2]
second seed	0.941	0.883	+8.3 [5.6, 11.0]	6.7	0.887	0.875	+2.0 [-0.8, 4.7]	+0.4 [-1.0, 2.3]	0.945	0.931	+4.4 [2.2, 7.5]	+0.3 [-0.7, 1.8]
same data, forged head only	0.949	0.896	+7.0 [4.7, 9.0]	7.4	0.914	0.883	+9.8 [7.0, 12.2]	-0.4 [-1.8, 0.9]	0.961	0.952	+4.2 [2.3, 6.6]	+0.2 [-1.0, 1.5]

TABLE III

WHERE THE DAMAGE COMES FROM. FF++-ONLY CLIP MODEL, FF++ VALIDATION FRAMES UNDER THE ELEVEN HELD-OUT FILTERS: COSINE OF THE MEAN FEATURE SHIFT WITH THE FORGED WEIGHT VECTOR w, MEAN CHANGE OF THE FORGED LOGIT ON GENUINE AND FORGED FRAMES, AND SHARE OF THE SHIFT ENERGY INSIDE THE 16-DIMENSIONAL SUBSPACE SPANNED BY TRAINING-TIME BEAUTIFICATIONS.

Held-out filter	n	$\cos(w, \bar{\Delta})$	Δlogit genuine	Δlogit forged	energy in V_{16}
earlybird	88	0.20	+7.84	+1.10	0.86
mayfair	94	0.33	+5.64	+1.80	0.83
kelvin	92	0.04	+2.40	-0.71	0.90
clarendon	99	0.07	+1.90	+0.40	0.86
lofi	95	0.06	+1.86	+0.17	0.87
moon	89	0.06	+1.75	-0.10	0.75
perpetua	107	0.09	+0.99	+0.21	0.77
inkwell	95	0.09	+0.86	-0.25	0.63
reyes	104	0.02	+0.76	-0.27	0.89
slumber	101	0.00	+0.43	-0.28	0.87
brooklyn	95	0.00	+0.14	-0.04	0.88

to 0.931) while its miss rate rises by 8.9 points, because the filters shift its scores for genuine and forged frames alike. We report both numbers for every model.

B. Where the damage comes from

We take the FF++-only CLIP model, whose forgery decision is a linear head w on the pooled feature, and measure how each held-out filter moves the feature of a validation frame (Table III, Fig. 3). The two filters with a radial luminance overlay, *earlybird* and *mayfair*, move the forgery logit of a genuine frame by $+7.8$ and $+5.6$; the nine pure colour looks move it by $+0.1$ to $+2.4$. The mean shifts of the two overlay filters have cosines of 0.20 and 0.33 with w, those of the colour looks 0.00 to 0.09, and the shift of *earlybird* is as long as the feature itself. Forged frames move far less (-0.7 to $+1.8$). A luminance ramp around the face reads as forgery evidence, and it does so for a genuine face far more than for a forged one, which already carries that evidence.

The shifts are not new directions: for every held-out filter 63–90% of the shift energy lies in the subspace spanned by the

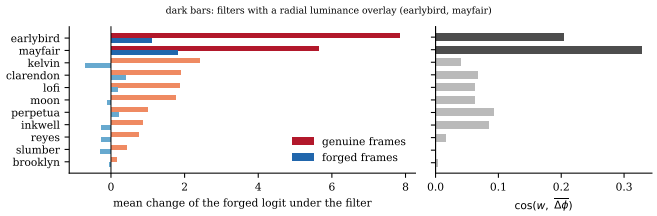


Fig. 3. Mean change of the forged logit under each held-out filter (left) and cosine of the mean feature shift with the forged weight vector (right), FF++-only CLIP model on FF++ validation frames. Dark bars: the two filters with a radial luminance overlay.

TABLE IV

WHICH CLASS RECEIVES BEAUTIFIED TRAINING VIEWS. FF++ ONLY, FORGED HEAD ONLY, EVERYTHING ELSE IDENTICAL; THREE SEEDS. CELEB-DF-B UNDER THE ONE RULE. P1 IS RELIABLY THE WORST; P2 IS ON AVERAGE THE BEST.

Arm	Beautified training views	AUC_b (3 seeds)	$\Delta\text{MISS}_{\text{beauty}}$ (3 seeds)	FA_b
P0	no beautified views	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	5.1 / 5.6 / 6.8
P1	beautified genuine faces only	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	4.5 / 3.9 / 3.7
P2	beautified genuine and forged faces	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	5.6 / 5.3 / 5.4
Paired differences, 95% video-cluster intervals (seeds 1 / 2 / 3)				
P1 − P0	$\Delta\text{MISS}_{\text{beauty}}$	+4.5 [1.5, 7.6] / +7.5 [4.4, 10.8] / +9.5 [6.7, 12.3]		
P2 − P0	$\Delta\text{MISS}_{\text{beauty}}$	-7.2 [-9.4, -5.0] / -4.0 [-6.1, -2.0] / +2.0 [0.1, 4.0]		
P2 − P1	$\Delta\text{MISS}_{\text{beauty}}$	-11.7 [-14.5, -8.8] / -11.5 [-14.5, -8.9] / -7.5 [-10.0, -5.0]		

shifts that training-time beautifications produce. This suggests a linear fix, projecting the subspace out of the feature and refitting the head, and it fails. Removing two directions costs the clean AUC at once (0.987 to 0.933) and the false-alarm rate under the held-out filters does not fall (17.5–27.5% against 21.5%), because the directions that beautification moves along carry the forgery evidence (supplement). The separation has to be learnt during training.

C. Exposure decides the sign of the error

With the architecture, data and recipe held fixed, we vary only which class receives beautified training views (Table IV): none (P0), genuine faces only (P1), or genuine and forged

faces (P2). P1 is the worst arm in every seed. Beautifying only genuine faces teaches the detector that beautification means genuine, and beautified deepfakes then escape it 4.5–9.5 points more often than they escape the detector that never saw beautification (every paired interval above zero), and 7.5–11.7 points more often than the detector that saw it on both classes. On the Celeb-DF-v2 filters P1's miss rate rises by 24 points, the largest of any model we trained. P2 against P0 is -7.2, -4.0 and $+2.0$ points of miss rise over the three seeds and $+0.020$, $+0.010$ and -0.006 of AUC under the presets: better on average, within the seed-to-seed spread of P0 itself (14.0 ± 3.1 points). On the two colour-filter sources the effect is consistent: P2 lowers the miss rise by 5–10 points in 9 of 10 seed-by-source cells (intervals below zero) and raises the false-alarm rise by 2–5 points, while the AUC under the filters does not change (supplement, Table S4). Exposure therefore moves a detector's beautification errors between misses and false alarms; it does not make genuine and forged faces more separable. Untreated AUC on Celeb-DF-B is the same for the three arms; on Celeb-DF-v2 P2 costs 0.012–0.021. The ordering replicates when Effort's own tuning scheme, the SVD residual of every self-attention projection [3] (0.21 M parameters), replaces LoRA inside the same pipeline: -4.6 and -5.2 points of miss rise for both-class over no exposure in two seeds, intervals excluding zero. This is the mechanism behind the sign of the errors in Table II: a detector's beautification error is set by what its training showed it, not by its backbone. Published detectors show the same thing without our intervention: Effort and GenD [5] adapt the same CLIP backbone, and under the Celeb-DF-v2 filters Effort's misses rise by 23.3 points with fewer false alarms while GenD's false alarms rise by 9.9 points with fewer misses. We also note that the no-exposure CLIP model already keeps more AUC under the presets (0.860–0.883) than Effort's released checkpoint (0.807), so most of the margin of our models over the two CLIP baselines comes from the base recipe (self-blends, degradation shared across the triplet, selection on validation) rather than from exposure.

D. What did not work

Three families of methods were tried under pre-registered bars and are reported because the bars were not met. (i) Adversarial training over a measured beautification operator: a differentiable operator over tone curves, colour matrix, saturation, contrast, a smooth luminance and tint field, smoothing and grain, with ranges fitted to the Tencent and Megvii renders and to app filters (it reproduces app filters at 40 dB PSNR and explains 7 dB of a commercial render), and training on its worst-case edit found by signed-gradient ascent on both classes. One-step and three-step search, a search confined to the range of measured edits, a consistency objective, random edits and the worst of four random draws all end within ±0.01 AUC of random colour grading under beautification and lose 0.02–0.05 untreated AUC on Celeb-DF-v2. The search itself works (genuine accuracy under the operator rises from 8.5% to 71–77%); its robustness does not transfer to real presets, and the label loss on edited views, dominated by the far easier attacks on genuine faces, teaches "heavily edited means less likely forged" (forged accuracy under random edits 89.5 to 79.5%). (ii) A learned inverse of the edit in front of the detector: the exact algebraic inverse restores a synthetic edit to 53–58 dB when the parameters are known, but a network cannot recover them from one image (26.7 dB), and the module lowers nothing. (iii) A decision pooled from the top cells of the footprint-supervised evidence map: it halves the false accusations under filters but lets 7.5 points more beautified forgeries through, because concentrated local evidence is also what the presets wash out. A third output class, finally, is not a detection contribution: the binary control is 0.02 AUC better on every source. Details and bars are in the supplement.

E. Commercial retouching

Three-way models trained on FF++ alone separate retouched from untouched faces at chance (balanced accuracy 48–55%) and call commercial smoothing fake, up to 86% of faces at the strongest level; the eleven published detectors trained on FF++ do the same, 36–63% of the most strongly smoothed faces at a threshold that accuses 5% of the originals (Table V). Trained with the single-operation components of two other services, OURS calls 0.6% of the retouched faces fake and 0.8% at the strongest smoothing, sends 67.7% of them to FILTER (smoothing at level 90: 98.8%) and gives 92.9% of renders a higher filter score than their own original. It also sends 20.8% of the untouched originals to FILTER, so its balanced accuracy on "retouched or not" is 73.5% (75.4% for the second seed), against our pre-registered target of 75%. The AUROC of $P(\text{filter})$ between originals and renders is 0.80–0.82 for every model trained with components, so no threshold reaches the target with margin: a single light operation moves the score by about one standard deviation of the spread between photographs.

F. Forgery detection under each detector's own preprocessing

Our frames are cropped by one pipeline (landmark box with a 35% margin), while the ten DeepfakeBench-trained baselines were trained on dlib-aligned 256 px crops. We therefore also score each baseline on crops of its own pipeline (supplement, Table S1): Forensics Adapter then reproduces its paper almost exactly (Celeb-DF-v2 frame AUC 0.899, video 0.956, against published 0.900 and 0.957), Effort does not (0.879 and 0.927 against 0.901 and 0.956, consistent with the re-run of its released weights by others [5]), and the eight CNNs gain 0.00–0.05. Compared each under its own crop, with paired intervals over the common videos, OURS is on par with both CLIP detectors on untreated Celeb-DF-v2 (frame 0.891; vs Effort $[-0.013, 0.038]$, vs Forensics Adapter $[-0.030, 0.013]$) and DFD (0.942; $[-0.006, 0.025]$ and $[-0.013, 0.017]$), at video level 0.947 and 0.970. The binary control is 0.914 and 0.967, above Effort with intervals excluding zero and on par with Forensics Adapter; a model trained on FF++ alone with our synthesised part-level forgeries does not reach it, so the margin of the full-data models comes from the external real data

TABLE V

RETOUCHING BY A SERVICE NEVER SEEN IN TRAINING: 2,210 FFHQ ORIGINALS AND THEIR ALIBABA RENDERS (FOUR OPERATIONS AT THREE LEVELS, 26,519 IMAGES). PAIRED RANK: SHARE OF PAIRS IN WHICH THE RENDER GETS A HIGHER $P(\text{filter})$ THAN ITS OWN ORIGINAL (50 MEANS NO SIGNAL). NATIVE THREE-WAY RULE. ALL VARIANTS IN THE SUPPLEMENT.

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, observed fakes (SUP)	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Trained with two vendors' renders and their single-operation components; Alibaba never seen (ours)							
OURS (CLIP ViT-L/14 + LoRA)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
OURS-lite (EfficientNet-B4)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4

(FFHQ stills and commercial renders), not from the synthesis. Under the app presets OURS keeps an AUC of 0.866 and 0.883 (two seeds) against 0.807 and 0.828, and its beautification-specific miss rise is 6.7 and 8.3 points against 11.0 and 15.2.

G. Localisation of part-level forgeries

For a detected part-level forgery with footprint M, let $\hat{A}$ be the $|M|$ highest cells of $\hat{m}$ and $x_A = x \odot (1 - A) + x_0 \odot A$ restore a region A to the original. On the three held-out editors the evidence head of OURS reaches IoU_7 0.83–0.84 with the true footprint, and restoring the region it names removes 93.5–94.0% of its detections, against 93.6–94.1% when the true footprint is restored (Table VI, Fig. 4); the second seed gives 90.2–90.9% against 90.2–91.0%. Restoring the complement removes at most 0.4%, a centred region 39–41% and a random region of equal area 10–16%. The SDXL editor, not seen in training, behaves like the others. The named region is therefore sufficient and necessary for the model's own decision on these forgeries. Post-hoc maps on a backbone without the head reach IoU_7 0.33–0.41 and flips of 52–57%, and the same model trained without part-level forgeries 0.38–0.48 and 54–79%; the footprint supervision carries the result. For whole-face swaps no method finds a small sufficient region.

VI. LIMITATIONS

The measurement covers four app presets, 27 colour filters and one retouching service; AR filters that re-render the face were excluded because their capture pipeline leaks the label (a model recognised them with the face greyed out). Celeb-DF-v2 under filters was used to diagnose the mechanism and is reported as a development source; Celeb-DF-B and the DFD filters were read once per model. The exposure effect is reliable in sign and on the miss rate; the gain of both-class over no exposure is within seed noise, and our models' margin over the CLIP baselines under presets is mostly the base recipe. Separating lightly retouched faces from their originals stays at 73–75% on an unseen service. The evidence head localises part-level forgeries, not whole-face swaps. DFDC, DFDCP and DF40 are not evaluated, and our DFD set is 332 of its videos, so no DFD number of ours is comparable to a published one; only detectors run by us on the same videos are compared. The published numbers of Effort were not reproduced by us or by others.

VII. CONCLUSION

Under one rule, beautification makes each of fifteen published face-forgery detectors accuse beautified genuine faces or let beautified deepfakes through, and the four CLIP detectors of 2025–26 lose 0.08–0.12 AUC under app presets. The damage has a measurable mechanism, a luminance ramp around the face read as forgery evidence along directions shared with real forgery evidence, and a measurable cause: a detector learns beautification as a label of whichever class it was shown on. Showing it on both classes, with commercial retouching split into single operations and part-level forgeries with exact footprints, gives a CLIP detector that is on par with Effort and Forensics Adapter on untreated data, keeps 0.04–0.07 more AUC under app presets, does not accuse retouched faces, and names the regions its decisions depend on. Adversarial training over a measured beautification operator, a learned inverse and a map-pooled decision did not improve on this, and we report them so that the next attempt starts further along.

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TABLE VI

EXPLANATIONS ON HELD-OUT PART-LEVEL FORGERIES (FF++ TEST FRAMES AND TEST-VIDEO DONORS, DETECTED ITEMS ONLY). IoU_7: IOU WITH THE EXACT FOOTPRINT ON A 7×7 GRID. FLIP: SHARE OF ITEMS THAT LEAVE *FAKE* WHEN THE NAMED REGION, OF THE SAME AREA AS THE FOOTPRINT, IS RESTORED TO THE ORIGINAL. CEIL.: THE SAME FOR THE TRUE FOOTPRINT. COMPL.: THE SAME FOR THE COMPLEMENT. SDXL WAS NEVER USED IN TRAINING.

	donor transplant				SD-1.5 inpaint				SDXL inpaint (unseen editor)			
Map	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.
centre prior / random region (last col.)	0.310	38.6	–	17.0	0.249	40.9	–	17.9	0.255	40.2	–	17.7
Score-CAM, same backbone, no head	0.408	56.5	85.8	5.9	0.334	52.4	85.2	15.4	0.341	52.4	85.9	17.1
head, vendor data, no part edits (CLIP)	0.479	78.8	89.4	2.5	0.376	53.9	69.1	14.4	0.445	68.9	78.3	8.4
head, part edits, FF++ only (EffNet-B4)	0.723	84.0	84.7	0.4	0.752	82.6	83.2	0.7	0.746	82.9	83.8	0.7
OURS-lite (EffNet-B4)	0.807	87.9	88.0	0.0	0.820	87.4	87.6	0.4	0.812	86.9	87.2	0.4
OURS (CLIP ViT-L/14 + LoRA)	0.834	94.0	94.1	0.1	0.841	93.6	93.7	0.4	0.831	93.5	93.6	0.3

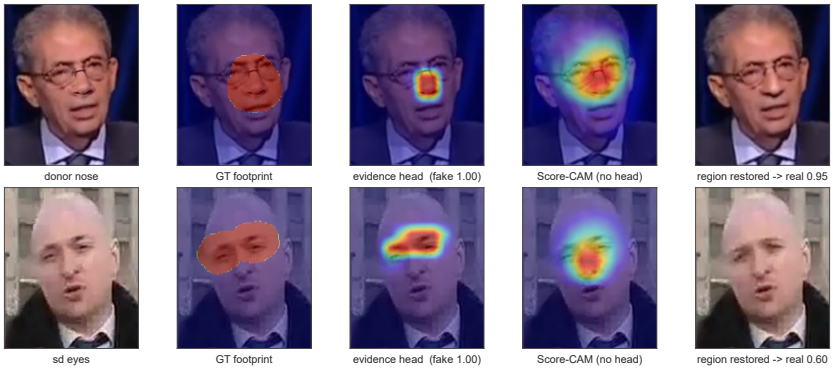


Fig. 4. Held-out part-level forgeries: input, exact footprint, evidence map with the three-way decision, Score-CAM on the same backbone without the head, and the result of restoring the named region to the original.

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